

No. of Functional Features Included in the Solution

Date	12 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input	Accepts soil and weather data	UI	Done	High
2	Input Validation	Validates entered values	Backend	Done	High
3	Crop Prediction	Predicts suitable crop	ML Model	Done	High
4	Confidence Score	Shows prediction confidence	Backend	Done	Medium
5	User Interface	Displays prediction result	Frontend	Done	Medium
6	Dataset Loading	Loads crop dataset	Backend	Done	Medium
7	Model Integration	Loads trained model	Backend	Done	High
8	Deployment	Hosted using Render	Deployment	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Crop prediction, Input validation, ML model, User interface

Additional / Nice-to-Have Features	Confidence score, Responsive design
Features Tested & Verified	All implemented features tested successfully.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input form, Result page
2	Backend / Logic	3	Validation, Prediction, Processing
3	Database / Storage	1	Crop dataset
4	API / Integration	1	Flask routing
5	Security / Authentication	1	Input validation